

Introduction to Capstone Project: Scenario and Review Criteria

Estimated time: 4 minutes

A national car dealership with local branches spread across the United States recently conducted a market survey. One of the suggestions that emerged from the survey was that customers would find it beneficial if they could access a central database of dealership reviews across the country.

You are a new hire at the company. You are assigned the task of building a website that allows new and existing customers to look up different branches by state and look at customer reviews of the various branches. Customers should be able to create an account and add their review for any of the branches. The management hopes this will bring transparency to the system and also increase the trust customers have in the dealership.

After thorough research and brainstorming, the team developed use cases for anonymous, authorized, and admin users.

Use cases for anonymous users

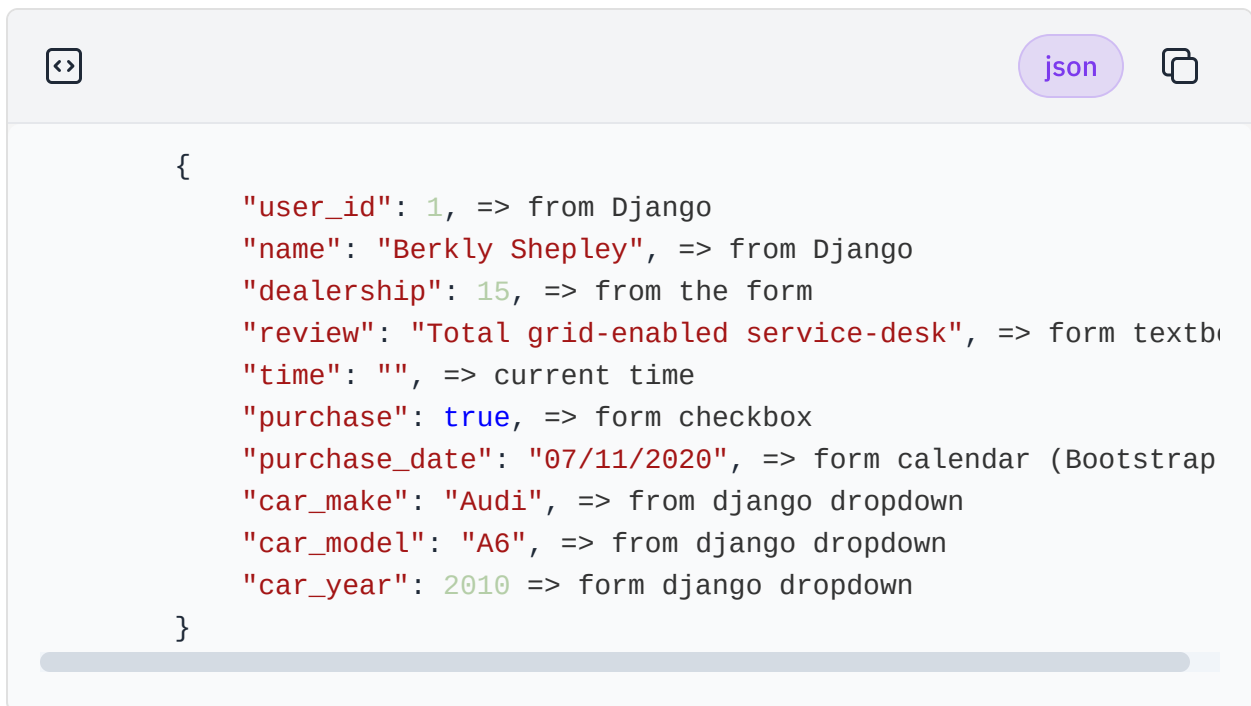
- View the **Contact Us** page
- View the **About Us** page
- View the list of dealerships
- Filter the list of dealerships by state:
 - Select **Show all** or a specific state from the State dropdown on the dealership page.
 - View all states if nothing is selected in the dropdown.

- Select a dealership to view the reviews for that dealership on the details page with each review displayed on a Bootstrap card
- Log in using their credentials.

Use cases for authorized users

In addition to the above, authorized users should be able to write a review for any dealership on the dealership's page. In order to enable authorized users to write their reviews:

- You can view a **Review** button against each dealer listed in the dealership table
- Selecting the **Review** button should take the user to the review page
- Fill out the form on the review page and submit it. This will add the review



```
{
  "user_id": 1, => from Django
  "name": "Berkly Shepley", => from Django
  "dealership": 15, => from the form
  "review": "Total grid-enabled service-desk", => form textb
  "time": "", => current time
  "purchase": true, => form checkbox
  "purchase_date": "07/11/2020", => form calendar (Bootstrap
  "car_make": "Audi", => from django dropdown
  "car_model": "A6", => from django dropdown
  "car_year": 2010 => form django dropdown
}
```

- Submitting the review will bring you back to the dealership detail page with the submitted review featured at the top of the reviews list, sorted by time.

Use cases for admin users

- Log in to the admin site with a predefined username and password
- Add new make, model, and other attributes

Your organization has assigned you as the Lead Full-Stack Software Developer on this project. Your job is to develop this portal as part of your Capstone project by following best practices for Full-Stack software development.

Review criteria (38 points)

The capstone project is divided into five modules. Each module has a quiz followed by a final submission that is graded by your peers in this course. The grading is divided as follows:

- Checklist: Static Pages (5 points)
- Checklist: User Management (6 points)
- Checklist: Back End (12 points)
- Checklist: Dynamic Pages (6 points)
- Checklist: CI & Containerize (9 points)

Evaluation criteria (50 Points)

You can submit your project deliverables in one of the following ways:

Option 1: AI-Graded Submission and Evaluation When you choose Option 1, you will be redirected to an AI tool where you can upload your deliverables, which may include URLs, terminal outputs, code snippets, or screenshots. You will then receive an AI-generated grade that will automatically reflect on your progress page.

Option 2: Peer-Graded Submission and Evaluation When you choose Option 2, you will upload your deliverables—such as URLs, terminal outputs, code snippets, or screenshots—through the My Submission section. Your submission will then be reviewed either by your peers or by an AI grader.

We recommend using Option 1 for faster grading. However, if you face any issues or cannot access it, you may use Option 2 instead.

If you encounter any grading problems, please reach out to the Course Team through the Discussion Forums

Please find the details of the Grading Criteria below:

► **Criteria for Option 1: AI-Graded Submission and Evaluation:**

► **Criteria for Option 2: Peer-Graded Submission and Evaluation:**

Next Steps

Be sure to read the capstone overview before starting with the step-by-step instructions.

Author(s)

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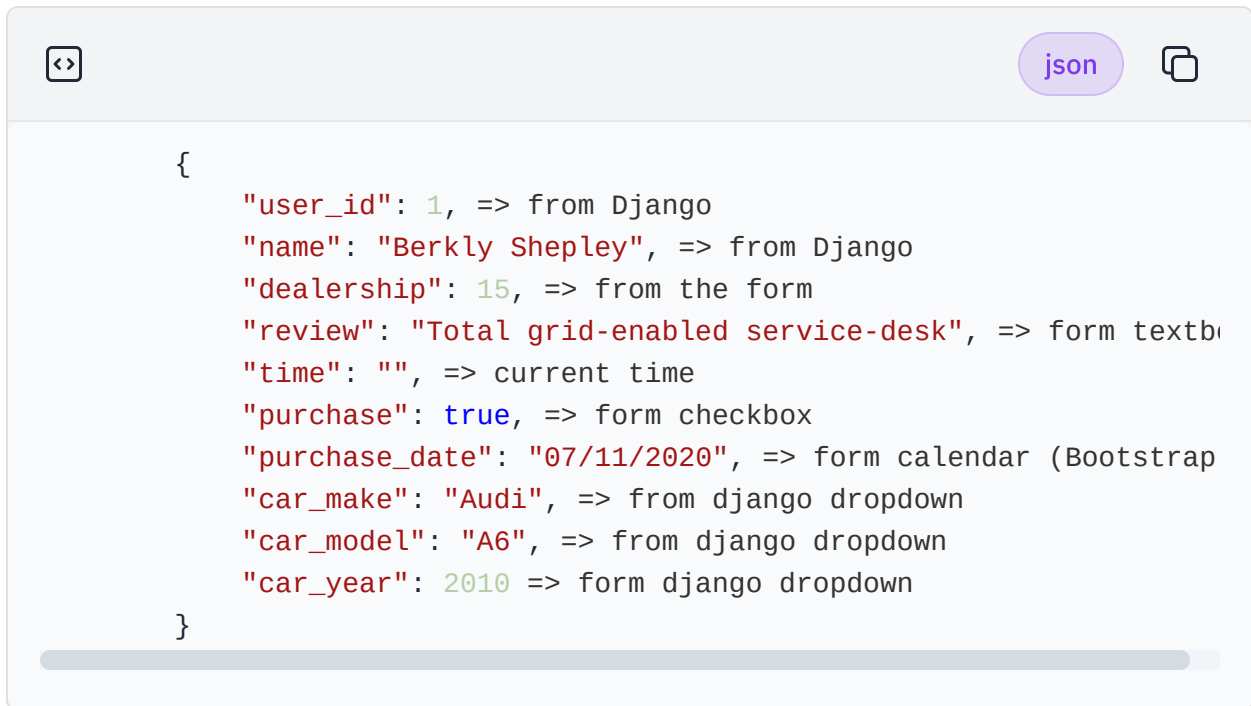
Use cases for anonymous users

- View the **Contact Us** page
- View the **About Us** page
- View the list of dealerships
- Filter the list of dealerships by state:
 - Select **Show all** or a specific state from the State dropdown on the dealership page.
 - View all states if nothing is selected in the dropdown.
 - View a table of dealerships for the selected state when the form is submitted.
- Select a dealership to view the reviews for that dealership on the details page with each review displayed on a Bootstrap card
- Log in using their credentials.

Use cases for authorized users

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We recommend using Option 1 for faster grading. However, if you face any issues or cannot access it, you may use Option 2 instead.

If you encounter any grading problems, please reach out to the Course Team through the Discussion Forums.

Please find the details of the Grading Criteria below:

▼ Criteria for Option 1: AI-Graded Submission and Evaluation:

Task 1: Update the `README.md` in your GitHub repository with the project name and all required project details. – **1 point**

Task 2: Run your Django development server and verify that it starts without errors. – **1 point**

Task 3: Update `About.html` with valid CSS links and realistic content including images, names, roles, descriptions, and email IDs. – **3 points**

Task 4: Update `Contact.html` with working CSS links, active navigation, required images, and complete contact details. – **2 points**

Task 5: Test the login endpoint using valid credentials and ensure the request succeeds. – **2 points**

Task 6: Test the logout endpoint and confirm that the logout operation succeeds. – **2 points**

Task 7: Build the Sign-Up page (`Register.jsx`) with fields for Username, First Name, Last Name, Email, Password, and a working Register button. – **1 points**

Task 8: Retrieve review data for any dealer ID using the correct endpoint and confirm that the expected reviews load. – **2 points**

Task 9: Retrieve the list of all dealers using the backend endpoint and confirm the returned data is correct. – **2 points**

Task 10: Retrieve information for a specific dealer using a dealer ID and confirm accuracy of the response. – **2 points**

Task 11: Retrieve dealers located in Kansas using the appropriate endpoint and verify the filtered results. – **2 points**

Task 12: Log in to the Django admin with the root user and confirm access is successful. – **2 points**

Task 13: Log out from the Django admin page and confirm the logout action works.

– **1 point**

Task 14: Retrieve car makes using the backend endpoint and confirm the list displays properly. – **2 points**

Task 15: Retrieve car models using the backend endpoint and confirm the list displays properly. – **2 points**

Task 16: Run sentiment analysis for the text "Fantastic services" and verify that the result includes the correct sentiment. – **2 points**

Task 17: Open the Django home page before logging in and confirm that the dealers list displays correctly. – **1 point**

Task 18: Log in to the Django application and verify the dealers page shows your username and the Review Dealer option. – **2 points**

Task 19: Filter dealers by state in the UI and verify that the results update based on the selected state. – **2 points**

Task 20: Open the dealer details page and confirm that dealer information and reviews load correctly. – **1 point**

Task 21: Open the Add Review page and ensure the full form is displayed before submitting a review. – **1 point**

Task 22: Submit a new review and verify that it appears correctly in the review list. – **2 points**

Task 23: Run your GitHub Actions workflow and confirm that all steps execute successfully. – **3 points**

Task 24: Deploy your Django application and ensure that the deployment loads without issues. – **1 point**

Task 25: Open the deployed landing page and verify that it loads correctly with the expected content. – **2 points**

Task 26: Log in to the deployed site and confirm that your username appears on

Task 27: Open the deployed dealer details page and verify that all dealer data displays correctly. – **2 points**

Task 28: Verify that a newly added review appears correctly in the deployed application. – **2 points**

► **Criteria for Option 2: Peer-Graded Submission and Evaluation:**

Next Steps

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